



Course Outline

Department of Electrical & Computer Engineering

School of Engineering and Physical Sciences

North South University, Bashundhara R/A, Dhaka – 1229, Bangladesh

1. Course No: CSE331L
2. Course Title: Microprocessor Interfacing and Embedded Systems Lab
3. Prerequisites: CSE332
4. Contact hour: 3 hours/week
5. Course Instructor: Md. Arifur Rahman (MARh)
6. Lab Instructor: Sheikh Mohammed Wali Ullah
Email: sheikh.ullah@northsouth.edu
7. Office Room: Not declared yet
8. Office hours: RA 1:00 PM – 6:30 PM
(A prior appointment is mandatory before visiting the office)
9. Class Room: LIB 609
10. Class Time: R 8:00 AM – 11:10 AM
11. Course Summary: This course provides an introduction to the fundamental concept of microprocessor interfacing and microprocessor based embedded systems. A basic idea of the internal and external architecture of a modern microprocessor will be studied followed by the interface mechanisms for such microprocessors. The course will also cover the common peripheral devices for microprocessor based systems. The course will then cover the programming languages for interfacing such as C or Assembly language that implements interrupt, data conversion algorithms, and inter-chip communications. A brief introduction to a modern Microcontroller will also be provided. Software Development Kits (SDKs) for embedded systems will be introduced in the laboratory classes for performing both simulation-based and hardware-based projects. This course has separate mandatory laboratory sessions every week as CSE 331L.
12. Course Objectives: The objectives of this course are –
 - a. to introduce the internal and external architecture of a modern microprocessor
 - b. to explain the interconnection of microprocessor and different peripheral devices

- c. to introduce Assembly language for direct manipulation of a microprocessor
- d. to introduce to SDKs for embedded systems for both simulation and hardware based works

13. Course Outcomes:

Upon Successful completion of this course, students will be able to:

Sl.	CO Description	Weightage (%)
1	identify the building blocks of microprocessor based systems	20%
2	construct code in Assembly language for microprocessors and microcontrollers	30%
3	Use Software Development Kits (SDK) to program embedded systems	30%
4	Design microprocessor or microcontroller-based embedded systems using hardware and software interface	20%

14. Software Resources:

CPUlator Web based Simulator (<https://cpulator.01xz.net/>)

STM CubeMX1.12.1

Windows Command-line or Ubuntu Bash

Keil uVision 5.0

15. Weightage Distribution among Assessment Tools: (Tentative)

Assignment / Reports	10%
Attendance	10%
Lab Assessment	15%
Midterm	20%
Project	20%
Final	25%

16. Grading Policy:

As per NSU grading policy available in

<http://www.northsouth.edu/academic/grading-policy.html>

Tentative Lecture Plan: (may change)

Class	Date	Content
1	29-May	Introduction + Basics about ARM Architecture
2	5-Jun	No classes
3	12-Jun	No classes
4	19-Jun	Arm Assembly 1
5	26-Jun	Arm Assembly 2
6	3-Jul	Arm Assembly 3
7	10-Jul	Arm Assembly 4
8	17-Jul	Arm Assembly 5
9	24-Jul	Midterm
10	31-Jul	STM32 Introduction + Software Installation
11	7-Aug	STM32 + LEDs control
12	14-Aug	STM32 + LCD/OLED Control
13	21-Aug	Final Exam

Policy:

1. Plagiarism in any of the assessment will be graded as zero.
2. Cheating in Exam Hall is strictly prohibited. If found, student will be expelled.
3. There will be no makeup for midterm and final exam.
4. Lab Reports must be submitted in the next class as per given manual.